5B322 Medical Science I
1301 Catherine St.
Ann Arbor, MI 48109-2218
apboyle@umich.edu
boylelab.org

Alan P. Boyle

Education

2005–2009 **Doctor of Philosophy,** Computational Biology and Bioinformatics
Duke University, Durham, NC

2001–2005 **Bachelor of Science,** *summa cum laude,* Biochemistry and Molecular Biology
Bachelor of Science, *summa cum laude,* Computer Science
Mississippi State University, Starkville, MS

Academic Appointments

2025–present **Professor with tenure,** Department of Computational Medicine & Bioinformatics
Professor, Department of Human Genetics
University of Michigan, Ann Arbor, MI

2020–2025 **Associate Professor with tenure,** Department of Computational Medicine & Bioinformatics
Associate Professor, Department of Human Genetics
University of Michigan, Ann Arbor, MI

2015–2020 **Assistant Professor,** Department of Human Genetics
2014–2020 **Assistant Professor,** Department of Computational Medicine & Bioinformatics
University of Michigan, Ann Arbor, MI

2010–2014 **Postdoctoral Scholar,** Genetics
Stanford University, Stanford, CA; Advisor: Dr. Michael Snyder

Spring 2010 **Postdoctoral Associate,** Computational Biology
Duke University, Durham, NC; Advisor: Dr. Terrence S. Furey

Institutional Affiliations

2023–present **Core Member,** Rogel Cancer Center
2021–present **Affiliate,** Michigan Neuroscience Institute
2016–present **Member,** Center for RNA Biomedicine
2020–2023 **Affiliate Member,** Rogel Cancer Center

Other Professional Experience

2013–2014 **Consultant,** Color Genomics
Personalized medicine and genomics startup

Scholarships, Fellowships, and Honors

2022 Valuing our Own Award, Michigan Medicine
2019 Endowment for the Basic Sciences Teaching Award
2018 First Place in CAGI5 Regulation Saturation Challenge
2017 NSF CAREER Award
2016 Institutional nominee for W.M. Keck Foundation Medical Science Research Program
2016 Institutional nominee for Searle Scholar Award
2015–2017 Alfred P. Sloan Foundation Fellowship in Computational & Evolutionary Molecular Biology
2013–2014 NIH Pathway to Independence Award (K99/R00) [1K99HG007356-01]
2012 AAAS/Science Program for Excellence in Science
2005–2008 NSF Graduate Research Fellowship
2005–2009 James B. Duke Fellowship

Summer 2004	Mayo Clinic Summer Undergraduate Research Fellow
2003	Barry M. Goldwater Memorial Scholarship
Summer 2003	The Institute for Genomic Research (TIGR) Summer Fellow
2001	Robert C. Byrd Honors Scholarship
2001	Mississippi State University Presidential Scholarship
2001	National Merit Scholarship

Research Support

Active: Principal Investigator and Multi-Principal Investigator

2026–2031 R35 GM163661, NIH/NIGMS (PI: Boyle)
The role of mobile elements in chromatin regulation
Explores chromatin conformation and gene regulation changed induced by polymorphic transposable elements.

2026–2031 R01 NS145291, NIH/NINDS (Multi-PI: Mills, Boyle)
Characterization and functional impact of somatic numtogenesis in the human cortex
Characterizes somatic nuclear mitochondrial DNA insertions in the human cortex and tests their effects on gene expression, chromatin structure, aging, and brain function.

2026–2027 National Ataxia Foundation Research Grant (Co-PI: Burmeister, Boyle)
FGF14 intronic GAA repeats in vestibular problems beyond SCA27B
Examines genetic features of the FGF14 repeat that may underlie subclinical or atypical expansions.

2023–2028 UG3/UH3 NS132084, NIH/Office of the Director (Multi-PI: Mills, Boyle)
Molecular and Computational Tools for Identifying Somatic Mosaicism in Human Tissues
Develops long-read molecular and computational methods to identify somatic mosaicism in normal human tissues as part of the SMaHT Consortium.

2021–2027 U01 HG011952, NIH/NHGRI (PI: Boyle)
Predicting the impact of genomic variation on cellular states
Develops models and tools that use single-cell data to predict the effects of genomic variation as part of the IGVF Consortium.

Active: Co-Investigator and Consultant

2024–2029 R01 DE032699, NIH/NIDCR (Multi-PI: Brenner, Mills, Spector; Co-I with effort)
Defining the Role of HPV Integration Structures in HNSCC Molecular Heterogeneity

2023–2028 R01 NS099280, NIH/NINDS (PI: Todd; Consultant)
Hexanucleotide repeat translation in ALS and Frontotemporal Dementia

2022–2027 R01 CA260677, NIH/NCI (PI: Malek; Co-I with effort)
The Biology of Mutant STAT6 in Follicular Lymphoma

2022–2026 R01 NS122165, NIH/NINDS (PI: Castro; Co-I with effort)
Uncover the role of H3.3-G343R mutation in shaping the DNA damage response, anti-tumor immunity, and mechanisms of resistance in glioma

Completed: Principal Investigator and Multi-Principal Investigator

2022–2026 R01 GM144484, NIH/NIGMS (PI: Boyle)
Mobile element derived chromatin looping variability in human populations

2022–2025 R21 CA257864, NIH/NCI (PI: Boyle)
High-throughput inverted reporter assay for characterization of silencers and enhancer blockers

2022–2025 Taubman Institute Innovation Program, University of Michigan (Co-PI: Todd, Boyle, Mills)
Repetitive elements in human health and disease

2017–2025 | U41/U24 HG009293, NIH/NHGRI (Multi-PI: Boyle, Cherry)
RegulomeDB: A Resource for the Human Regulome

2020–2023 | R21 HG011493, NIH/NHGRI (Multi-PI: Boyle, Mills)
New technologies for accurate capture and sequencing of repeat-associated regions

2021–2023 | Cancer Center Discovery Award, University of Michigan (PI: Boyle)
Direct capture of complete HPV integration sites using long-read sequencing

2022 | R21 HG011493-02S1, NIH/NIA (Multi-PI: Boyle, Mills)
New technologies for accurate capture and sequencing of repeat-associated regions (supplement)

2022 | NVIDIA GPU Grant, NVIDIA Corporation (PI: Boyle)

2017–2022 | DBI-1651614, NSF/BIO/DBI (PI: Boyle)
CAREER: Conservation of cohesin-containing cis-regulatory modules in the human and mouse lineages

2019–2020 | Precision Health Investigators Award, University of Michigan (Co-PI: Todd, Boyle, Mills)
Short Tandem Repeats in Precision Health and Human Disease

2019 | NVIDIA GPU Grant, NVIDIA Corporation (PI: Boyle)

2017–2018 | Eleanor and Larry Jackier U-M/Technion and Weizmann Collaborative Research Grant (Co-PI: Boyle, Mandel-Gutfreund)
Michigan-Israel Partnership for Research and Education
Identifying novel disease-related mutations in the genomic environments around transcription factor binding sites

2015–2017 | FG-2015-65465, Alfred P. Sloan Foundation (PI: Boyle)
Sloan Research Fellowship in Computational and Evolutionary Molecular Biology

2013–2017 | R00 HG007356 Pathway to Independence Award (K99/R00), NIH/NHGRI (PI: Boyle)
Global Discovery and Validation of Functional Regulatory Elements

Completed: Co-Investigator and Consultant

2018–2026 | R01 HD093570, NIH/NICHD (PI: Bielas; Co-I with effort)
Genetic Diagnosis of Neurodevelopmental Disorders in India

2021–2026 | K08 HL153799, NIH/NHLBI (PI: Denstaedt; Consultant)
Predisposition for Lung Injury in Sepsis Survival

2021–2025 | R01 HD104680, NIH/NICHD (PI: Hammoud; Co-I with effort)
Sperm Chromatin: Implications on organismal development and fertility

2023–2025 | Michigan Alzheimer's Disease Center Developmental Project, University of Michigan (PI: Zhou; Consultant)
Explore the functional impact of transposable elements in Alzheimer's disease and related dementias

2017–2024 | R35 HL135824, NIH/NHLBI (PI: Willer; Co-I with effort)
Using Genetics to Inform Mechanism of Cardiovascular Disease

2020–2024 | W81XWH2010336, U.S. Department of Defense/Army (PI: Aguilar; Co-I with effort)
Understanding and Enhancing the Regenerative Capacity of Skeletal Muscle to Trauma by Targeting Muscle-Nerve Synergy

2016–2020 | R01 HL130705, NIH/NHLBI (PI: Willer; Co-I with effort)
Large-scale human genetics to understand molecular mechanisms of atrial fibrillation and related traits

Professional Service

Institutional and Consortium Leadership

2026–present	Michigan Neuroscience Institute Executive Committee
2022–present	University of Michigan Biomedical Research Council [Standing Member; Co-Chair 2026–present]
2014–2018, 2025–present	DCMB Admissions Committee [Chair 2025–present]
2024–present	Department of Human Genetics Faculty Development Committee
2023–present	R01 Bootcamp Medical School Cohort Coach
2023–present	Somatic Mosaicism across Human Tissues (SMaHT) Consortium Steering Committee
2021–present	Impact of Genomic Variation on Function (IGVF) Consortium Steering Committee
2018–present	DCMB Laboratory Safety Liaison
2015, 2016, 2025, 2026	DCMB Retreat Planning Committee Chair, including the first annual retreat
2020–2022	Department of Human Genetics M.S. Admissions Committee
2017–2024	DCMB Preliminary Exam Abstract Review Committee [Chair 2018–2022]
2019–2020	Department of Human Genetics Ph.D. Admissions Committee
2017–2020	Department of Human Genetics Faculty Recruitment and Promotions Committee
2016–2020	DCMB Seminar Series Committee [Chair]
2018–2019	Cellular and Molecular Biology Admissions Committee
2017–2019	Endowment for the Basic Sciences Faculty IT Committee
2016–2019	DCMB Faculty Recruitment Committee
2015–2017	Department of Human Genetics Computational Support Committee
2008–2009	Duke Computational Biology and Bioinformatics Student Committee

Editorial and Program Committee Service

2026	Academic Editor, PLOS ONE
2026	Program Committee, Proceedings, ISMB/ECCB
2023	Program Committee, Genome Sequence Analysis, ISMB/ECCB
2023	Program Committee, Biomedical Informatics, ISMB/ECCB
2018, 2020	Program Committee, Comparative and Functional Genomics, ISMB/ECCB
2018, 2019	Program Committee, Studies of Phenotypes and Clinical Applications, ISMB/ECCB
2019	Program Committee, General Computational Biology, ISMB/ECCB
2017	Program Committee, Regulatory Genomics Special Interest Group Meeting (RegGenSIG), ISMB/ECCB
2015–2018	Program Committee, Great Lakes Bioinformatics and Canadian Computational Biology Conference (GLBIO/CCBC)
2015–2016	Program Committee, Algorithms for Computational Biology (ALCOB)
2013–2016	Program Committee, Gene Regulation and Transcriptomics, ISMB/ECCB
2012–2015	DNA Day Essay Contest Detailed Review Judge, American Society of Human Genetics

Grant Review

2026	NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)
2026	NIH Study Section ZRG1 IIDB Z (70) - Stage I New Innovator Award Program (mail review)
2026	NIH Study Section ZRG1 MGG-F (90) S - Special Topics in Genomics, Genetic Evolution, and Technology Development (ad hoc)
2025	NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)
2024	NIH Study Section ZRG1 BBBT-D (50) - PAR Panel: Biodata Management and Common Fund Data Sets (ad hoc; Co-Chair)
2024	NIH Study Section BDMA - Biodata Management and Analysis Study Section (ad hoc)
2023	NSF Review Panel - Molecular and Cellular Biosciences, Genetic Mechanisms (ad hoc)
2023	NIH Study Section - Multi-Omics of Health and Disease Data Analysis and Coordination Center
2023	NIH Study Section GVE - Genetic Variation and Evolution Study Section (ad hoc)
2022	NIH Study Section ZRG1 ISB-S (57) - Academic-Industrial Partnerships for Translation of Technologies for Diagnosis and Treatment
2022	NASA Study Section E.11 Space Biology: Animal Studies - Omics Systems [21SBAS-OmicsSys] (ad hoc)

2020	NIH/NIMH Study Section ZMH1 ERB-C (08) - Fine-Mapping Genome-Wide Associated Loci to Identify Proximate Causal Mechanisms of Serious Mental Illness
2019	NIH/NIMH Study Section ZMH1 ERB-C (01) - PsychENCODE: Noncoding Functional Elements in the Human Brain and Their Role in the Development of Psychiatric Disorders
2018–2019	University of Michigan internal review, Searle Scholars Program
2015	UK Medical Research Council Methodology Research Panel (ad hoc)
2015	UK Biotechnology and Biological Sciences Research Council (ad hoc)
2015	Michigan Institute for Clinical and Health Research Postdoctoral Translational Scholars Program (ad hoc)

Manuscript Review

2009–present	*Ad hoc* reviewer with more than 100 verified reviews for journals including *Science*, *Nature Biotechnology*, *Nature Genetics*, *Genome Research*, *Genome Biology*, *Nature Communications*, *Bioinformatics*, *Nucleic Acids Research*, and *PLOS Computational Biology*
2012	Distinguished contributor as a leading reviewer for *Bioinformatics*

Professional Memberships

2018–present	American Society of Human Genetics
2013–present	International Society for Computational Biology
2012–present	American Association for the Advancement of Science
2005–present	Gamma Sigma Delta Agricultural Honor Society

Teaching and Mentorship

Training Programs

2024–present	**Member,** Systems & Integrative Biology Training Grant (SIB)
2021–present	**Member,** Biomedical Informatics and Data Science Training Program (BIDS-TP)
2017–present	**Member,** Cellular and Molecular Biology Program
2015–present	**Member,** Genome Science Training Program (GSTP)
2015–present	**Member,** Michigan Predoctoral Training Program in Genetics (GTP)
2014–present	**Member,** Program in Biomedical Sciences
2014–present	**Member,** Bioinformatics Training Program

Teaching (F = fall, W = winter, S = summer)

W19, W20, W21, W22, W23, W24, W25, W26	Bioinformatics Concepts and Algorithms (BIOINF 529) [Course Director]
F15, F16, F17, F18, F19, F20, F21	Gene Structure and Regulation (HUMGEN 541) [3 lectures + 2 discussions / yr.]
F19, F22	Research Responsibility and Ethics (PIBS 503) [1 discussion / yr.]
F21, W22, F24, W25	Genetics Student Seminar (HUMGEN 821/822) [Mentor]
F17, F18	Experimental Genetics Systems (HUMGEN 632) [Course Director]
F15, W16, F16, W17, F17, W18, F18	Bioinformatics Journal Club (BIOINF 602/603) [Course Director F18]
S17, S18	Introduction to Biocomputing Bootcamp (BIOSTAT/BIOINF/HUMGEN 606) [2 full days / yr.]
F15, F16, F17	Introduction to Bioinformatics & Computational Biology (BIOINF 527) [2 lectures + 3 labs / yr.]
S15, S16, S17	Basic Biology for Graduate Students with Quantitative Training (BIOINF 523) [2 lectures / yr.]
F03	Lab TA for Isotopes Tech I (Mississippi State University, BCH 4414)

Guest Lectures / Panels

2018–2019	Lecturer, REU Site: Mathematical and Theoretical Biology Institute (MTBI), Arizona State University (NSF1757968) [2 days]
2017	Panel member, University of Michigan “New Faculty Orientation to Corporate & Foundation Relations” [70 attendees]

2016 Experimental Genetics Systems (HUMGEN 632) [1 discussion]
2014 Panel member, BIOINF 527 “Challenges in Biology, Biomedicine, Data & Analysis”
2010 Co-taught Cold Spring Harbor Systems Biology Pre-meeting Workshop
2009 Duke student panelist for “How to prepare for and get into graduate school”
2008 Taught Duke mini-course on Genome Browsers & Databases

Mentorship

Ph.D. Students (n=21)

2026–present Kobe Howcroft (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Biomedical Informatics and Data Science Training Program (T32)

2024–present Ingrid Flaspohler (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Biomedical Informatics and Data Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2024–present Steve Losh (Ph.D. Student, Bioinformatics, University of Michigan)

2024–present Sowmya Srinivasan (Ph.D. Student, Genetics and Genomics, University of Michigan)
Career Training in Reproductive Biology (T32)

2022–present Katarina Pavlovic (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2022–present Rintsen Sherpa (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2021–2025 Kinsey Van Deynze (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Genome Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2020–2025 Andrea Valenzuela (Ph.D. Student, Chemical Biology, University of Michigan)
NIH Cellular Biotechnology Training Program (T32)

2020–2026 Breanna McBean (Ph.D. Student, Genetics and Genomics, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Genome Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2020–2025 Camille Mumm (Ph.D. Student, Genetics and Genomics, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Genome Science Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)
Rackham Predoctoral Fellowship

2018–2024 Bradley Crone (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2017–2023 Melissa Englund (Ph.D. Student, Genetics and Genomics, University of Michigan)
NIH Human Genetics Training Program (T32)
Rackham Graduate Student Research Grant (candidate)

2018–2023 Nanxiang (Samuel) Zhao (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2016–2018 Haley Amemiya (Ph.D. Student, Cellular and Molecular Biology, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan

NIH Cellular & Molecular Biology Training Program (T32)
NIH Cellular Biotechnology Training Program (T32) (Declined)
PIBS Excellence in Service Award
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)
Maas Professional Development Award
Rackham Graduate School Scholar-Activist Award

2016–2020 Shriya Sethuraman (Ph.D. Student, Bioinformatics, University of Michigan)

2016–2023 Christopher Castro (Ph.D. Student, Bioinformatics, University of Michigan)
NIH Bioinformatics Training Program (T32)
Rackham Merit Fellow
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)
Global Research Engagement Opportunity Fellowship

2017–2022 Ningxin Ouyang (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2016–2021 Shengcheng Dong (Ph.D. Student, Bioinformatics, University of Michigan)
Rackham Graduate Student Research Grant (candidate)

2015–2021 Torrin McDonald (Ph.D. Student, Genetics and Genomics, University of Michigan)
NIH Human Genetics Training Program (T32)
Rackham Graduate Student Research Grant (pre-candidate)
Rackham Graduate Student Research Grant (candidate)

2015–2017 Greg Farnum (Ph.D. Student, Cellular and Molecular Biology, University of Michigan)

2015–2020 Sierra Nishizaki (Ph.D. Student, Genetics and Genomics, University of Michigan)
Joint M.S. in Bioinformatics, University of Michigan
NIH Genome Science Training Program (T32)
Rackham Merit Fellow
Rackham Summer Award
Rackham Graduate Student Research Grant (candidate)

M.S. Students (n=8)

2026–present Lara Kok (M.S. Student, Genetics and Genomics, University of Michigan)
2025 Christine Geng (M.S. Student, Bioinformatics, University of Michigan)
2023 Hawra Aljawad (M.S. Student, Chemical Engineering, University of Michigan)
2023 Xinyi Liu (M.S. Student, Bioinformatics, University of Michigan)
2022–2023 Emily Pogson (M.S. Student, Genetics and Genomics, University of Michigan)
2019–2020 Monica Holmes (M.S. Student, Bioinformatics, University of Michigan)
2017–2018 Nanxiang (Samuel) Zhao (M.S. Student, Bioinformatics, University of Michigan)
2015–2017 Ningxin Ouyang (M.S. Student, Bioinformatics, University of Michigan)

Additional Graduate Rotation Students (n=17)

2024 Karan Smith (Rotation Student, Cellular and Molecular Biology, University of Michigan)
2024 Jun Sik Yun (Rotation Student, Genetics and Genomics, University of Michigan)
2024 Jeremy Chen (Rotation Student, Bioinformatics, University of Michigan)
2023 Rosina Carr (Rotation Student, Bioinformatics, University of Michigan)
2023 Connor Ward (Rotation Student, Medical Scientist Training Program, University of Michigan)
2022 Brandt Bessell (Rotation Student, Bioinformatics, University of Michigan)
2022 Xiaomeng Du (Rotation Student, Bioinformatics, University of Michigan)
2022 Mahnoor Gondal (Rotation Student, Bioinformatics, University of Michigan)
2022 Xin Li (Rotation Student, Biological Chemistry, University of Michigan)
2022 Bohan Chen (Rotation Student, Cell and Developmental Biology, University of Michigan)

2021 Amelia Lauth (Rotation Student, Cellular and Molecular Biology, University of Michigan)
2019 Margarita Brovkina (Rotation Student, Cellular and Molecular Biology, University of Michigan)
2018 Steve Ho (Rotation Student, Human Genetics, University of Michigan)
2018 Matthew Pun (Rotation Student, Medical Scientist Training Program, University of Michigan)
2017 Amanda Moccia (Rotation Student, Human Genetics, University of Michigan)
2017 Stephen Carney (Rotation Student, Human Genetics, University of Michigan)
2016 Tingyang Li (Rotation Student, Bioinformatics, University of Michigan)

Postdoctoral Fellows (n=4)

2026–present Kinsey Van Deynze (University of Michigan)
2023–2025 Melissa Englund (University of Michigan)
2022–2025 Torrin McDonald (University of Michigan)
2021–2022 Shengcheng Dong (University of Michigan)

Non-student Lab Volunteers (n=2)

2019–2021 Greg Farnum (University of Michigan)
2018–2019 Monica Holmes (Postbac, University of Michigan)

Undergraduate and High School Students (n=28)

2026 Rose Mills (High School)
2026–present Pria Chandarana (Undergraduate, Neuroscience, University of Michigan)
2026–present Cecelia Crandall (Undergraduate, Cognitive Science, University of Michigan)
2026 Boutheine Teyeb (Undergraduate, Computer Science, University of Michigan)
2026 Aanshi Agrawal (Undergraduate, Molecular, Cellular, and Developmental Biology, University of Michigan)
2025–2026 Krrish Thakker (Undergraduate, Biochemistry, University of Michigan)
2023–2024 Kateri Darr (Undergraduate, Computer Science, University of Michigan)
2023 Mason Miller (Undergraduate, Computer Science, University of Michigan)
2022–2024 Summer Ann (Undergraduate, Neuroscience, University of Michigan)
2022–2025 Kobe Howcroft (Undergraduate, Computer Science, University of Michigan)
2021–2024 Preston Parana (Undergraduate, UROP Molecular, Cellular, and Developmental Biology, University of Michigan)
UROP Blue Ribbon Award
2021–2022 Julia Tweadey (Undergraduate, LSA Honors Program, Life Science Informatics, University of Michigan)
2021 Aryn Booker (Undergraduate, UROP Molecular, Cellular, and Developmental Biology, University of Michigan)
UROP Blue Ribbon Award
2020 Marcela Alcaide Aligio (Undergraduate, SROP, Hunter College CUNY)
2019–2020 David Wang (Undergraduate, UROP Computer Science, University of Michigan)
2019–2020 Jack Lu (Undergraduate, UROP Computer Science, University of Michigan)
2019–2020 Diana Davis (Undergraduate, Neuroscience and German, University of Michigan)
2019 Sheila Rasouli (Undergraduate, Neuroscience, University of Toronto)
2019 Vibhasri Davuluri (High School, Girls Who Code Summer Intern)
2016–2019 Cody Morterud (Undergraduate, UROP Computer Science / Honors Capstone, University of Michigan)
2016–2017 Colten Williams (Undergraduate, UROP Computer Science, University of Michigan)
2016–2017 Courtney Asman (Undergraduate, Neuroscience, University of Michigan)
2014–2017 Maxwell Spadafore (Undergraduate, LS&A Honors Informatics, University of Michigan)
2013–2014 Natalie Ng (High School, Stanford Institutes of Medicine Summer Research)
2013–2014 Dana Wyman (Undergraduate, Biology, Stanford University)
2013 Justin Young (High School, Stanford Institutes of Medicine Summer Research)
2012 Melanie Connick (Undergraduate, Biology, University of New Mexico)
2012 Edward Dai (Undergraduate, Computer Science, Stanford University)

Doctoral Thesis Committees (n=53)

2026–present	Connor Davison (Genetics and Genomics, University of Michigan, Committee Member)
2026–present	Kobe Howcroft (Bioinformatics, University of Michigan, Chair)
2025–present	Brandt Bessell (Bioinformatics, University of Michigan, Committee Member)
2024–present	Weizhou Qian (Bioinformatics, University of Michigan, Committee Member)
2024–present	Jinhao Wang (Bioinformatics, University of Michigan, Committee Member)
2024–present	Brandon Klein (Medicinal Chemistry, University of Michigan, Committee Member)
2024–present	Bohan Chen (Bioinformatics, University of Michigan, Committee Member)
2024–present	Sowmya Srinivasan (Genetics and Genomics, University of Michigan, Co-Chair)
2024–present	Ingrid Flaspohler (Bioinformatics, University of Michigan, Chair)
2024–present	Steve Losh (Bioinformatics, University of Michigan, Chair)
2024–present	Lu Lu (Bioinformatics, University of Michigan, Committee Member)
2023–present	Linghua Jiang (Bioinformatics, University of Michigan, Committee Member)
2023–present	Elysia Chou (Bioinformatics, University of Michigan, Committee Member)
2023–present	Rebecca McAvoy (Molecular, Cellular, and Developmental Biology, University of Michigan, Committee Member)
2023–present	Chinmay Raut (Bioinformatics, University of Michigan, Committee Member)
2022–present	Katarina Pavlovic (Bioinformatics, University of Michigan, Chair)
2022–present	Rintsen Sherpa (Bioinformatics, University of Michigan, Chair)
2022–present	Emily Peirent (Neuroscience, University of Michigan, Committee Member)
2025–present	Monica Holmes (Bioinformatics, University of Michigan, Committee Member) *Single-Molecule Long-read Sequencing for Resolution of Structurally Complex Genomic Loci and Accurate Phenotype Prediction*
2024–2026	Matthew Hodgman (Bioinformatics, University of Michigan, Committee Member) *Raising the Standard of Machine Learning Transparency and Utility in Cardiovascular Medicine*
2020–2026	Breanna McBean (Genetics and Genomics, University of Michigan, Co-Chair) *Targeting Resistance and Metastasis: Advancing Local and Systemic Therapeutic Strategies for Aggressive Breast Cancers*
2021–2026	Mashiat Rabbani (Genetics and Genomics, University of Michigan, Committee Member) *A Genome-scale Program for Building the Sperm Epigenome*
2024–2025	Lingrui Cai (Bioinformatics, University of Michigan, Committee Member) *Advancing Medical Image and Report Analysis with Artificial Intelligence: Applications in Inflammatory Bowel Disease and Traumatic Brain Injury*
2021–2025	Kinsey Van Deynze (Bioinformatics, University of Michigan, Chair) *Sequence Analysis of Tandem Repeats Utilizing 3rd Generation Sequencing Technologies*
2020–2025	Andrea Valenzuela (Chemical Biology, University of Michigan, Co-Chair) *AI-Driven Workflows Integrating Strain Diversity and Host Metabolic-Genetic Variation for Tuberculosis Treatment Prioritization*
2021–2025	Zhenhao Zhang (Bioinformatics, University of Michigan, Committee Member) *Decoding Genome Regulation through Deep Learning Models*
2020–2025	Steve Ho (Genetics and Genomics, University of Michigan, Committee Member) *Applied Machine Learning for Big Data Genomics*
2020–2025	Camille Mumm (Genetics and Genomics, University of Michigan, Chair) *Exploring Repetitive Elements in Neurodegenerative Disease Using Targeted Long-Read Sequencing*
2022–2025	Kaiwen Deng (Bioinformatics, University of Michigan, Committee Member) *Advancing Computational Models and Algorithms for the Analysis of Single-Cell and Neural Data*
2022–2024	Franco Tavella (Biophysics, University of Michigan, Committee Member) *Robustness and Tunability of Biological Oscillations*
2018–2024	Bradley Crone (Bioinformatics, University of Michigan, Chair) *Computational Methods in Functional Prioritization of Polygenic Risk Score Models*
2021–2024	Wenjin Gu (Bioinformatics, University of Michigan, Committee Member) *Development of Viral Integration Analysis Technologies for Virus-Associated Cancer Research*
2018–2023	Rucheng Diao (Bioinformatics, University of Michigan, Committee Member) *Local Chromatin Environments Shape Transcription and Adaptive Immunity in Bacteria*
2021–2023	Zijun Gao (Bioinformatics, University of Michigan, Committee Member)

Advance Machine Learning and Image Analysis Methods for Clinical Decision Support in Cardiovascular and Pulmonary Diseases

2018–2023 Nanxiang (Samuel) Zhao (Bioinformatics, University of Michigan, Chair)
Decoding Regulatory Variants with Computational Methods in Non-coding Regions of the Human Genome

2020–2023 Ashley Melnick (Cellular and Molecular Biology, University of Michigan, Committee Member)
Cdc73 Protects Notch-Induced Leukemia Cells From DNA Damage and Mitochondrial Stress

2016–2023 Christopher Castro (Bioinformatics, University of Michigan, Chair)
Investigating the Role of Noncoding De Novo Single-Nucleotide Variants in Autism Spectrum Disorder

2017–2023 Melissa Englund (Genetics and Genomics, University of Michigan, Chair)
Identification and Characterization of Cis-Regulatory Elements in the Human Genome

2018–2023 Stephen Carney (Cancer Biology, University of Michigan, Committee Member)
Epigenetic reprogramming in mutant IDH1 glioma influences radioresistance and neural lineage differentiation

2019–2023 Benjamin Yang (Biomedical Engineering, University of Michigan, Committee Member)
Towards Defining Principles of Cell Fate Plasticity

2018–2022 Marcus Sherman (Bioinformatics, University of Michigan, Committee Member)
Cultivation of enhanced bioinformatic-specific pedagogical manipulatives, interventions, and professional development

2021–2022 Kuan-Han Hank Wu (Bioinformatics, University of Michigan, Committee Member)
Integrating Electronic Health Records with Genetic Information to Advance Precision Medicine Approaches in Cardiovascular Disease

2017–2022 Amanda Moccia (Genetics and Genomics, University of Michigan, Committee Member)
Investigation of Developmental Disorders: Genetic Discovery and Functional Validation

2017–2022 Ningxin Ouyang (Bioinformatics, University of Michigan, Chair)
Deciphering Transcriptional Regulatory Circuits: Transcription Factor Binding and Regulatory Variants Identification

2015–2021 Torrin McDonald (Genetics and Genomics, University of Michigan, Chair)
Leveraging New Technologies to Explore Regulatory and Structural Elements of the Human Genome

2018–2021 Heming Yao (Bioinformatics, University of Michigan, Committee Member)
Machine Learning and Image Processing for Clinical Outcome Prediction: Applications in Medical Data from Patients with Traumatic Brain Injury, Ulcerative Colitis, and Heart Failure

2016–2021 Mohd Hafiz Bin Mohd Rothi (Molecular, Cellular, and Developmental Biology, University of Michigan, Committee Member)
Control of Chromatin by RNA-mediated Transcriptional Silencing

2016–2021 Shengcheng Dong (Bioinformatics, University of Michigan, Chair)
Computational Methods to Identify Regulatory Variants in the Non-coding Regions of the Human Genome

2017–2021 Steven Romanelli (Molecular & Integrative Physiology, University of Michigan, Committee Member)
Viral CRISPR/Cas9 Gene Transfer for Somatic Knockout in Brown Adipose Tissue

2018–2021 Negar Farzaneh (Bioinformatics, University of Michigan, Committee Member)
Automated Decision Support System for Traumatic Injuries

2016–2020 Shriya Sethuraman (Bioinformatics, University of Michigan, Co-Chair)
Genome-wide Identification of Non-coding Transcription by RNA Polymerase V and Its Involvement in Transcriptional Gene Silencing

2015–2020 Sierra Nishizaki (Genetics and Genomics, University of Michigan, Chair)
Decoding the Non-coding Genome: Novel Technologies for the Characterization of Non-coding Elements and Variation

2017–2020 Christopher Lee (Biostatistics, University of Michigan, Committee Member)
Improvements and Developments in Gene Regulation and Single-Cell Gene Expression Data Analysis

2018–2019 Christine Ziegler (Biological Chemistry, University of Michigan, Committee Member)

2015–2018 Ari Allyn-Feuer (Bioinformatics, University of Michigan, Committee Member)

The Pharmacoepigenomics Informatics Pipeline and H-GREEN Hi-C Compiler: Discovering Pharmacogenomic Variants and Pathways with the Epigenome and Spatial Genome
2015–2017 Raymond Cavalcante (Bioinformatics, University of Michigan, Committee Member)
Beyond the Transcriptome: Facilitating Interpretation of Epigenomics and Metabolomics Data
2015–2017 Zhengting Zou (Bioinformatics, University of Michigan, Committee Member)
Model-based genomic studies of protein sequence evolution: convergence, epistasis, and amino acid acceptance rates

Preliminary Exam Committees (n=46)

2026 James George (Cellular and Molecular Biology, University of Michigan)
2025 Xinyi Deng (Bioinformatics, University of Michigan)
2025 Mitchell Witt (Bioinformatics, University of Michigan)
2024 Luke Gohmann (Cellular and Molecular Biology, University of Michigan)
2024 Benjamin Li (Bioinformatics, University of Michigan)
2024 Tiffany Wan (Bioinformatics, University of Michigan)
2024 Zhiyuan Yu (Bioinformatics, University of Michigan)
2024 Rebecca McAvoy (Molecular, Cellular, and Developmental Biology, University of Michigan)
2024 Bonje Obua (Cellular and Molecular Biology, University of Michigan)
2024 Abigail Vallie (Cellular and Molecular Biology, University of Michigan)
2023 Jinhao Wang (Bioinformatics, University of Michigan)
2023 Lishi Yin (Bioinformatics, University of Michigan)
2023 Matthew Hodgman (Bioinformatics, University of Michigan)
2023 Ilakkiya Venkatachalam (Genetics and Genomics, University of Michigan)
2023 Jianhui Gong (Bioinformatics, University of Michigan)
2023 Mahnoor Gondal (Bioinformatics, University of Michigan)
2023 Elysia Chou (Bioinformatics, University of Michigan)
2022 Sean Moran (Bioinformatics, University of Michigan)
2022 Lu Lu (Bioinformatics, University of Michigan)
2022 Linghua Jiang (Bioinformatics, University of Michigan)
2022 Kaiwen Deng (Bioinformatics, University of Michigan)
2022 Yufeng Zhang (Bioinformatics, University of Michigan)
2021 Anthony Nguyen (Human Genetics, University of Michigan)
2021 Hanbyul Cho (Bioinformatics, University of Michigan)
2021 Charles Ryan (Cellular and Molecular Biology, University of Michigan)
2021 Kuan-Han Wu (Bioinformatics, University of Michigan)
2021 Wenjin Gu (Bioinformatics, University of Michigan)
2020 Jie Cao (Bioinformatics, University of Michigan)
2020 Zijun Gao (Bioinformatics, University of Michigan)
2020 Ashley Melnick (Cellular and Molecular Biology, University of Michigan)
2019 Benjamin Yang (Biomedical Engineering, University of Michigan)
2019 Maria Virgilio (Cellular and Molecular Biology, University of Michigan)
2018 Zhi Carrie Li (Bioinformatics, University of Michigan)
2018 Kevin Hu (Bioinformatics, University of Michigan)
2018 Siyu Liu (Bioinformatics, University of Michigan)
2018 Alexandra Weber (Bioinformatics, University of Michigan)
2018 Mitch Fernandez (Bioinformatics, University of Michigan)
2017 Tingyang Li (Bioinformatics, University of Michigan)
2017 Marcus Sherman (Bioinformatics, University of Michigan)
2017 Adrienne Shami (Human Genetics, University of Michigan)
2017 Trenton Frisbie (Human Genetics, University of Michigan)
2017 Melissa Englund (Human Genetics, University of Michigan)
2017 Peter Orchard (Bioinformatics, University of Michigan)
2017 Li Guan (Bioinformatics, University of Michigan)
2016 Shriya Sethuraman (Bioinformatics, University of Michigan)
2016 Jed Carlson (Bioinformatics, University of Michigan)

Publications

* co-first authorship; † co-senior authorship; underline indicates Boyle Lab members

[89] The Somatic Mosaicism across Human Tissues Network (SMaHT). "Comprehensive benchmarking of somatic mutation detection by the SMaHT Network." *Cell* 2026, 189:6594–6612.e21. PMID: 42880231.

[88] Denstaedt SJ, McBean B, **Boyle AP**, Arenberg BC, Mack M, Moore BB, Newstead MW, Deng Y, Nesvizhskii AI, Singer BH, Cano J, Prescott HC, Goodridge HS, Zemans RL. "Long-term immune reprogramming of classical monocytes with altered ontogeny mediates enhanced lung injury in sepsis survivors." *JCI Insight* 2026. PMID: 40475424.

[87] Deng X, Du X, Gensterblum-Miller E, Currie J, Majchrowski BK, Lialios P, Bhangale A, Spector ME, **Boyle AP**, †Brenner JC, †Mills RE. "Challenges in the detection and assembly of virus integration structures in human genomes." *Frontiers in Oncology* 2026, 16:1925720. PMID: 42807781.

[86] The Somatic Mosaicism across Human Tissues Network. "Integrated map of somatic mosaicism across human tissues in 25 individuals." *bioRxiv* 2026. doi: 10.64898/2026.09.01.748636.

[85] Pavlovic K, McDonald TL, Diehl AG, Switzenberg JA, **Boyle AP**. "Fiber-TEnCATS reveals haplotype-specific chromatin accessibility and DNA methylation at human L1HS loci." *bioRxiv* 2026. PMID: 42395424.

[84] Kenney GE, Sherpa RN, Burgess JD, **Boyle AP**, Phanstiel DH. "SEMPLR: an R package for transcription factor binding prediction." *Bioinformatics* 2026, 42(6):btag383. PMID: 42286344.

[83] Diehl AG, **Boyle AP**. "GPatch enables chromosome-scale, gap-free pseudoassemblies from fragmented draft genomes." *bioRxiv* 2026. PMID: 40501647.

[82] Rabbani M, Apell Z, Parnell TJ, Moritz L, Kim S, Srinivasan S, Agrawal R, Vargo A, Orchard P, Xie W, Freddolino L, **Boyle AP**, Li JZ, Lesch BJ, Cairns B, Kim M, Wilson TE, Hammoud SS. "3D chromatin compartment of round spermatids encodes the spatiotemporal program of histone-to-protamine exchange in spermiogenesis." *bioRxiv* 2026. PMID: 41959395.

[81] Li S, Chou E, Wang K, **Boyle AP**, Sartor MA. "TFBSpedia: a comprehensive human and mouse transcription factor binding sites database." *bioRxiv* 2026. doi: 10.64898/2026.03.04.709638.

[80] *McBean B, *Hauk B, *Michmerhuizen AR, Chandler BC, Pesch AM, Lerner LM, Gurdak D, Ward C, Rana P, Zeidane RA, Mercer V, Tao M, Hochmuth E, Jungles KM, The S, Liu M, Spratt DE, **Boyle AP**, Pierce LJ, Speers CW. "Transcriptomic analysis to uncover the mechanism of radiosensitization of AR-positive triple-negative breast cancers with AR inhibition." *npj Breast Cancer* 2026, 12(1):50. PMID: 41735341.

[79] *Zhao N, *Sherpa RN, Dong S, Howcroft KB, **Boyle AP**. "Organ-specific prioritization and annotation of non-coding regulatory variants in the human genome." *bioRxiv* 2026. doi: 10.1101/2023.09.07.556700.

[78] Li D, Liu S, Assis PR, Li M, Dong S, Whaling I, Jolanki O, Kagda M, Zhang W, Macias-Velasco JF, Liu T, Cody S, Antonacci-Fulton L, Huang Y, Liu J, Montgomery MT, Zeiberg D, Jain S, Pejaver V, Bergquist T, Chen Y, Radivojac P, Gersbach CA, Sherpa RN, Castro CP, **Boyle AP**, Starita LM, Fowler DM, Ahituv N, Dey KK, Majoros WH, Reddy TE, Craven M, Sinha R, Sverchkov Y, Cai X, Nzima MZ, Calderwood MA, Rozowsky J, Gerstein M, Ma J, Yue F, Cherry JM, Love MI, Engreitz JM, Hitz BC, Wang T. "The IGVF catalog—from genetic variation to function." *Nucleic Acids Research* 2026, 54(D1):D1437–D1445. PMID: 41359121.

[77] *Zhou W, *Mumm C, *Gan Y, Switzenberg JA, Wang J, De Oliveira P, Kathuria K, Losh SJ, Bessell B, McDonald TL, †McConnell MJ, †**Boyle AP**, †Mills RE. "A personalized multi-platform assessment of somatic mosaicism in the human frontal cortex." *bioRxiv* 2025. PMID: 39763954.

[76] *Wang S, *Bae M, *Wang J, Zhao B, Nguyen K, Mallett S, Switzenberg JA, Losh SJ, Sexton CE, Miao B, Dong S, Zeng X, Wang Z, McDonald TL, Mumm C, Gadde RK, Tariq AM, Chen Z, Feng WC, Burn A, Park J, Chu C, Shen H, Wang T, Urban AE, Zhu X, Li H, Burns KH, Chun HJE, Park PJ, SMaHT MEI Working Group, †**Boyle AP**, †Mills RE, †Zhou W, †Lee EA. "Multi-platform framework for mapping somatic retrotransposition in human tissues." *Cell Genomics* 2025, in press. PMID: 41278868.

[75] Miller SL, Green KM, Crone B, Switzenberg JA, Tank EMH, Krans A, Jansen-West K, Wieland CM, Ji EW, Petrucelli L, Barmada SJ, **Boyle AP**, Todd PK. “Cryptic intronic transcriptional initiation generates efficient endogenous mRNA templates for C9orf72-associated RAN translation.” *Proceedings of the National Academy of Sciences* 2025, 122(32):e2507334122. PMID: 40758885.

[74] Parana P, Mumm C, McConnell MJ, **Boyle AP**. “Draft de novo genome construction of Scytonema sp. PRP1: identified from single-cell sequencing library preparation.” *Microbiology Resource Announcements* 2025, 14(9):e00029-25. PMID: 40744872.

[73] Coorens THH, Oh JW, Choi YA, Lim NS, Zhao B, Voshall A, Abyzov A, Antonacci-Fulton L, Aparicio S, Ardlie KG, Bell TJ, Bennett JT, Bernstein BE, Blanchard TG, **Boyle AP**, Buenrostro JD, Burns KH, Chen F, Chen R, Choudhury S, Doddapaneni HV, Eichler EE, Evrony GD, Faith MA, Fazzio TG, Fulton RS, Garber M, Gehlenborg N, Germer S, Getz G, Gibbs RA, Hernandez RG, Jin F, Korbel JO, Landau DA, Lawson HA, Lennon NJ, Li H, Li Y, Loh PR, Marth G, McConnell MJ, Mills RE, Montgomery SB, Natarajan P, Park PJ, Satija R, Sedlazeck FJ, Shao DD, Shen H, Stergachis AB, Underhill HR, Urban AE, VonDran MW, Walsh CA, Wang T, Wu TP, Zong C, Lee EA, Vaccarino FM, Somatic Mosaicism across Human Tissues Network. “The Somatic Mosaicism across Human Tissues Network.” *Nature* 2025, 643(8070):47–59. PMID: 40604182.

[72] Lee S, McAfee JC, Lee J, Gomez A, Ledford AT, Clarke D, Min H, Gerstein MB, **Boyle AP**, Sullivan PF, Beltran AS, Won H. “Massively parallel reporter assay investigates shared genetic variants of eight psychiatric disorders.” *Cell* 2025, 188(5):1409–1424.e21. PMID: 39848247.

[71] McBean B, Abou Zeidane R, Lichtman-Mikol S, Hauk B, Speers J, Tidmore S, Flores CL, Rana PS, Pisano C, Liu M, Santola A, Montero A, **Boyle AP**, Speers CW. “MELK as a Mediator of Stemness and Metastasis in Aggressive Subtypes of Breast Cancer.” *International Journal of Molecular Sciences* 2025, 26(5):2245. PMID: 40076867.

[70] *Van Deynze K, *Mumm C, Maltby CJ, Switzenberg JA, Todd PK, **Boyle AP**. “Enhanced detection and genotyping of disease-associated tandem repeats using HMMSTR and targeted long-read sequencing.” *Nucleic Acids Research* 2025, 53(2):gkae1202. PMID: 39676678.

[69] Maltby CJ, Krans A, Grudzien SJ, Palacios Y, Muiños J, Suárez A, Asher M, Willey S, Van Deynze K, Mumm C, **Boyle AP**, Cortese A, Khurana V, Barmada SJ, Dijkstra AA, Todd PK. “AAGGG repeat expansions trigger RFC1-independent synaptic dysregulation in human CANVAS neurons.” *Science Advances* 2024, 10(36):eadn2321. PMID: 39231235.

[68] IGVF Consortium. “Deciphering the impact of genomic variation on function.” *Nature* 2024, 633(8028):47–57. PMID: 39232149.

[67] Crone B, **Boyle AP**. “Enhancing Portability of Trans-Ancestral Polygenic Risk Scores through Tissue-Specific Functional Genomic Data Integration.” *PLoS Genetics* 2024, 20(8):e1011356. PMID: 39110742.

[66] Yee C, Xiao Y, Chen H, Reddy A, Xu B, Medwig-Kinney T, Zhang W, **Boyle AP**, Herbst W, Xiang Y, Matus D, Shen K. “An activity-regulated transcriptional program directly drives synaptogenesis.” *Nature Neuroscience* 2024, 27(9):1695–1707. PMID: 39103556.

[65] Holmes MJ, Mahjour B, Castro CP, Farnum GA, Diehl AG, **Boyle AP**. “HaplotagLR: an efficient and configurable utility for haplotagging long reads.” *PLoS ONE* 2024, 19(3):e0298688. PMID: 38478504.

[64] The Critical Assessment of Genome Interpretation Consortium. “CAGI, the Critical Assessment of Genome Interpretation, establishes progress and prospects for computational genetic variant interpretation methods.” *Genome Biology* 2024, 25(1):53. PMID: 38389099.

[63] McAfee JC, Lee S, Lee J, Bell JL, Krupa O, Davis J, Insigne K, Bond ML, Zhao N, **Boyle AP**, Phanstiel DH, Love MI, Stein JL, Ruzicka WB, Davila-Velderrain J, Kosuri S, Won H. “Systematic investigation of allelic regulatory activity of schizophrenia-associated common variants.” *Cell Genomics* 2023, 3(10):100404. PMID: 37868037.

[62] Moritz L, Schon SB, Rabbani M, Sheng Y, Agrawal R, Glass-Klaiber J, Sultan C, M CJ, Clements J, Baldwin MR, Diehl AG, **Boyle AP**, O’Brien PJ, Ragunathan K, Hu YC, Kelleher NL, Nandakumar J, Li JZ, Orwig KE, Redding S, Hammoud SS. “Sperm chromatin structure and reproductive fitness are altered by substitution of a single amino acid in mouse protamine 1.” *Nature Structural & Molecular Biology* 2023, 30(8):1077–1091. PMID: 37460896.

[61] Castro CP, Diehl AG, **Boyle AP**. “Challenges in screening for de novo noncoding variants contributing to genetically complex phenotypes.” *Human Genetics and Genomics Advances* 2023, 4(3):100210. PMID: 37305558.

[60] *Mumm C, *Drexel ML, McDonald TL, Diehl AG, Switzenberg JA, **Boyle AP**. “OnRamp: rapid nanopore plasmid validation.” *Genome Research* 2023, 33(5):741–749. PMID: 37156622.

[59] *Dong S, *Zhao N, Spragins E, Kagda MS, Li M, Assis PR, Jolanki O, Luo Y, Cherry JM, †**Boyle AP**, †Hitz BC. “Annotating and prioritizing human non-coding variants with RegulomeDB v.2.” *Nature Genetics* 2023, 55(5):724–726. PMID: 37173523.

[58] Zhao N, Wang S, Huang Q, Dong S, **Boyle AP**. “Explain-seq: an end-to-end pipeline from training to interpretation of sequence-based deep learning models.” *bioRxiv* 2023. doi: 10.1101/2023.01.23.525250.

[57] Ouyang N, **Boyle AP**. “Quantitative assessment of association between noncoding variants and transcription factor binding.” *bioRxiv* 2022. doi: 10.1101/2022.11.22.517559.

[56] Nishizaki SS, **Boyle AP**. “SEMplMe: A tool for integrating DNA methylation effects in transcription factor binding affinity predictions.” *BMC Bioinformatics* 2022, 23(1):317. PMID: 35927613.

[55] Qin T, Lee C, Li S, Cavalcante RG, Orchard P, Yao H, Zhang H, Wang S, Patil S, **Boyle AP**, Sartor MA. “Comprehensive enhancer-target gene assignments improve gene set level interpretation of genome-wide regulatory data.” *Genome Biology* 2022, 23(1):105. PMID: 35473573.

[54] Dong S, **Boyle AP**. “Prioritization of regulatory variants with tissue-specific function in the non-coding regions of human genome.” *Nucleic Acids Research* 2022, 50(1):e6. PMID: 34648033.

[53] Bao Y, Wadden J, Erb-Downward JR, Ranjan P, Zhou W, McDonald TL, Mills RE, **Boyle AP**, Dickson RP, Blaauw D, Welch JD. “SquiggleNet: real-time, direct classification of nanopore signals.” *Genome Biology* 2021, 22(1):298. PMID: 34706748.

[52] *Nishizaki SS, *McDonald TL, Farnum GA, Holmes MJ, Drexel ML, Switzenberg JA, **Boyle AP**. “The inducible lac operator-repressor system is functional in zebrafish cells.” *Frontiers in Genetics* 2021, 12:683394. PMID: 34220959.

[51] *McDonald TL, *Zhou W, Castro CP, Mumm C, Switzenberg JA, †Mills RE, †**Boyle AP**. “Cas9 targeted enrichment of mobile elements using nanopore sequencing.” *Nature Communications* 2021, 12(1):3586. PMID: 34117247.

[50] Zhao N, **Boyle AP**. “F-Seq2: improving the feature density based peak caller with dynamic statistics.” *NAR Genomics and Bioinformatics* 2021, 3(1):lqab012. PMID: 33655209.

[49] *Tsuzuki M, *Sethuraman S, Coke AN, Rothi MH, **Boyle AP**, Wierzbicki AT. “Broad noncoding transcription suggests genome surveillance by RNA polymerase V.” *Proceedings of the National Academy of Sciences* 2020, 117(48):30799–30804. PMID: 33199612.

[48] Rothi MH, Sethuraman S, Dolata J, **Boyle AP**, Wierzbicki AT. “DNA methylation directs nucleosome positioning in RNA-mediated transcriptional silencing.” *bioRxiv* 2020. doi: 10.1101/2020.10.29.359794.

[47] Diehl AG, **Boyle AP**. “MapGL: Inferring evolutionary gain and loss of short genomic sequence features by phylogenetic maximum parsimony.” *BMC Bioinformatics* 2020, 21(1):416. PMID: 32962625.

[46] The ENCODE Project Consortium. “Perspectives on ENCODE.” *Nature* 2020, 583(7818):693–698. PMID: 32728248.

[45] The ENCODE Project Consortium. “Expanded encyclopaedias of DNA elements in the human and mouse genomes.” *Nature* 2020, 583(7818):699–710. PMID: 32728249.

[44] Ouyang N, **Boyle AP**. “TRACE: transcription factor footprinting using chromatin accessibility data and DNA sequence.” *Genome Research* 2020, 30(7):1040–1046. PMID: 32660981.

[43] Diehl AG, Ouyang N, **Boyle AP**. “Transposable elements contribute to cell and species-specific chromatin looping and gene regulation in mammalian genomes.” *Nature Communications* 2020, 11(1):1796. PMID: 32286261.

[42] Lee CT, Cavalcante RG, Lee C, Qin T, Patil S, Wang S, Tsai ZTY, **Boyle AP**, Sartor MA. “Poly-Enrich: count-based methods for gene set enrichment testing with genomic regions.” *NAR Genomics and Bioinformatics* 2020, 2(1):lqaa006. PMID: 32051932.

[41] Nishizaki SS, Ng N, Dong S, Porter RS, Morterud C, Williams C, Asman C, Switzenberg JA, **Boyle AP**. “Predicting the effects of SNPs on transcription factor binding affinity.” *Bioinformatics* 2020, 36(2):364–372. PMID: 31373606.

[40] Shigaki D, Adato O, Adhikar AN, Dong S, Hawkins-Hooker A, Inoue F, Juven-Gershon T, Kenlay H, Martin B, Patra A, Penzar DP, Schubach M, Xiong C, Yan Z, **Boyle AP**, Kreimer A, Kulakovskiy IV, Reid J, Unger R, Yosef N, Shendure J, Ahituv N, Kircher M, Beer MA. “Integration of Multiple Epigenomic Marks Improves Prediction of Variant Impact in Saturation Mutagenesis Reporter Assay.” *Human Mutation* 2019, 40(9):1280–1291. PMID: 31106481.

[39] Dong S, **Boyle AP**. “Predicting functional variants in enhancer and promoter elements using RegulomeDB.” *Human Mutation* 2019, 40(9):1292–1298. PMID: 31228310.

[38] Diehl AG, **Boyle AP**. “CGIMP: Real-time exploration and covariate projection for self-organizing map datasets.” *Journal of Open Source Software* 2019, 4(39):1520. PMID: 32500114.

[37] Amemiya HM, †Kundaje A, †**Boyle AP**. “The ENCODE Blacklist: Identification of Problematic Regions of the Genome.” *Scientific Reports* 2019, 9(1):9354. PMID: 31249361.

[36] Varshney A, VanRenterghem H, Orchard P, †**Boyle AP**, †Stitzel ML, †Ucar D, Parker SC. “Cell specificity of regulatory annotations and their genetic effects on gene expression.” *Genetics* 2019, 211(2):549–562. PMID: 30593493.

[35] Diehl AG, **Boyle AP**. “Conserved and species-specific transcription factor co-binding patterns drive divergent gene regulation in human and mouse.” *Nucleic Acids Research* 2018, 46(4):1878–1894. PMID: 29361190.

[34] Nielsen JB, Fritsche LG, Zhou W, Teslovich TM, Holmen OL, Gustafsson S, Gabrielsen ME, Schmidt EM, Beaumont R, Wolford BN, Lin M, Brummett CM, Preuss MH, Refsgaard L, Bottinger EP, Graham SE, Surakka I, Chu Y, Skogholt AH, Dalen H, **Boyle AP**, Oral H, Herron TJ, Kitzman J, Jalife J, Svendsen JH, Olesen MS, Njølstad I, Løchen ML, Baras A, Gottesman O, Marcketta A, O’Dushlaine C, Ritchie MD, Wilsgaard T, Loos RJF, Frayling TM, Boehnke M, Ingelsson E, Carey DJ, Dewey FE, Kang HM, Abecasis GR, Hveem K, Willer CJ. “Genome-wide Study of Atrial Fibrillation Identifies Seven Risk Loci and Highlights Biological Pathways and Regulatory Elements Involved in Cardiac Development.” *American Journal of Human Genetics* 2018, 102(1):103–115. PMID: 29290336.

[33] Spadafore M, Najarian K, **Boyle AP**. “A proximity-based graph clustering method for the identification and application of transcription factor clusters.” *BMC Bioinformatics* 2017, 18(1):530. PMID: 29187152.

[32] *Yang B, *Zhou W, *Jiao J, Nielsen JB, Mathis MR, Heydarpour M, Lettre G, Folkersen L, Prakash S, Schurmann C, Fritsche L, Farnum GA, Lin M, Othman M, Hornsby W, Driscoll A, Levasseur A, Thomas M, Farhat L, Dubé MP, Isselbacher EM, Franco-Cereceda A, Guo DC, Bottinger EP, Deeb GM, Booher A, Kheterpal S, Chen YE, Kang HM, Kitzman J, Cordell HJ, Keavney BD, Goodship JA, Ganesh SK, Abecasis G, Eagle KA, **Boyle AP**, Loos RJF, †Eriksson P, †Tardif JC, †Brummett CM, †Milewicz DM, †Body SC, †Willer CJ. “Protein-altering and regulatory genetic variants near GATA4 implicated in bicuspid aortic valve.” *Nature Communications* 2017, 8(1):15481. PMID: 28541271.

[31] Nishizaki SS, **Boyle AP**. “Mining the Unknown: Assigning Function to Noncoding Single Nucleotide Polymorphisms.” *Trends in Genetics* 2017, 33(1):34–45. PMID: 27939749.

[30] Diehl AG, **Boyle AP**. “Deciphering ENCODE.” *Trends in Genetics* 2016, 32(4):238–249. PMID: 26962025.

[29] Phanstiel DH, **Boyle AP**, Heidari N, Snyder MP. “Mango: A bias correcting ChIA-PET analysis pipeline.” *Bioinformatics* 2015, 31(19):3092–3098. PMID: 26034063.

[28] *Yue F, *Cheng Y, *Breschi A, *Vierstra J, *Wu W, *Ryba T, *Sandstrom R, *Ma Z, *Davis C, *Pope BD, *Shen Y, Pervouchine DD, Djebali S, Thurman RE, Kaul R, Rynes E, Kirilusha A, Marinov GK, Williams BA, Trout D, Amrhein H, Fisher-Aylor K, Antoshechkin I, DeSalvo G, See LH, Fastuca M, Drenkow J, Zaleski C, Dobin A, Prieto P, Lagarde J, Bussotti G, Tanzer A, Denas O, Li K, Bender MA, Zhang M, Byron R, Groudine MT,

McCleary D, Pham L, Ye Z, Kuan S, Edsall L, Wu YC, Rasmussen MD, Bansal MS, Kellis M, Keller CA, Morrissey CS, Mishra T, Jain D, Dogan N, Harris RS, Cayting P, Kawli T, **Boyle AP**, Euskirchen G, Kundaje A, Lin S, Lin Y, Jansen C, Malladi VS, Cline MS, Erickson DT, Kirkup VM, Learned K, Sloan CA, Rosenbloom KR, Lacerda de Sousa B, Beal K, Pignatelli M, Flicek P, Lian J, Kahveci T, Lee D, Kent WJ, Ramalho Santos M, Herrero J, Notredame C, Johnson A, Vong S, Lee K, Bates D, Neri F, Diegel M, Canfield T, Sabo PJ, Wilken MS, Reh TA, Giste E, Shafer A, Kutyavin T, Haugen E, Dunn D, Reynolds AP, Neph S, Humbert R, Hansen RS, De Bruijn M, Selleri L, Rudensky A, Josefowicz S, Samstein R, Eichler EE, Orkin SH, Levasseur D, Papayannopoulou T, Chang KH, Skoultchi A, Gosh S, Disteche C, Treuting P, Wang Y, Weiss MJ, Blobel GA, Cao X, Zhong S, Wang T, Good PJ, Lowdon RF, Adams LB, Zhou XQ, Pazin MJ, Feingold EA, Wold B, Taylor J, Mortazavi A, Weissman SM, Stamatoyannopoulos JA, Snyder MP, Guigo R, Gingeras TR, Gilbert DM, Hardison RC, Beer MA, Ren B, Mouse ENCODE Consortium. “A comparative encyclopedia of DNA elements in the mouse genome.” *Nature* 2014, 515(7527):355–364. PMID: 25409824.

[27] *Cheng Y, *Ma Z, Kim BH, Wu W, Cayting P, **Boyle AP**, Sundaram V, Xing X, Dogan N, Li J, Euskirchen G, Lin S, Lin Y, Visel A, Kawli T, Yang X, Patacsil D, Keller CA, Giardine B, Mouse ENCODE Consortium, Kundaje A, Wang T, Pennacchio LA, Weng Z, †Hardison RC, †Snyder MP. “Principles of regulatory information conservation between mouse and human.” *Nature* 2014, 515(7527):371–375. PMID: 25409826.

[26] Phanstiel DH, **Boyle AP**, Araya CL, Snyder MP. “Sushi.R: flexible, quantitative and integrative genomic visualizations for publication-quality multi-panel figures.” *Bioinformatics* 2014, 30(19):2808–2810. PMID: 24903420.

[25] ***Boyle AP**, *Araya CL, Brdlik C, Cayting P, Cheng C, Cheng Y, Gardner K, Hillier LW, Janette J, Jiang L, Kasper D, Kawli T, Kheradpour P, Kundaje A, Li JJ, Ma L, Niu W, Rehm EJ, Rozowsky J, Slattery M, Spokony R, Terrell R, Vafeados D, Wang D, Weisdepp P, Wu YC, Xie D, Yan KK, Feingold EA, Good PJ, Pazin MJ, Huang H, Bickel PJ, Brenner SE, Reinke V, Waterston RH, Gerstein M, †White KP, †Kellis M, †Snyder M. “Comparative analysis of regulatory information and circuits across distant species.” *Nature* 2014, 512(7515):453–456. PMID: 25164757.

[24] Araya CL, Kawli T, Kundaje A, Jiang L, Wu B, Vafeados D, Terrell R, Weissdepp P, Gevirtzman L, Mace D, Niu W, **Boyle AP**, Xie D, Ma L, Murray JI, Reinke V, Waterston RH, Snyder M. “Regulatory analysis of the C. elegans genome with spatiotemporal resolution.” *Nature* 2014, 512(7515):400–405. PMID: 25164749.

[23] *Kasowski M, *Kyriazopoulou-Panagiotopoulou S, *Grubert F, *Zaugg JB, *Kundaje A, Liu Y, **Boyle AP**, Zhang QC, Zakharia F, Spacek DV, Li J, Xie D, Steinmetz LM, Hogenesch JB, Kellis M, Batzoglou S, Snyder M. “Extensive variation in chromatin states across humans.” *Science* 2013, 342(6159):750–752. PMID: 24136358.

[22] *Xie D, ***Boyle AP**, *Wu L, Kawli T, Zhai J, Snyder M. “Dynamic trans-acting factor colocalization in human cells.” *Cell* 2013, 155(3):713–724. PMID: 24243024.

[21] Schaub MA, **Boyle AP**, Kundaje A, †Batzoglou S, †Snyder M. “Linking disease associations with regulatory information in the human genome.” *Genome Research* 2012, 22(9):1748–1759. PMID: 22955986.

[20] *Gerstein MB, *Kundaje A, *Hariharan M, *Landt SG, *Yan KK, *Cheng C, *Mu XJ, *Khurana E, *Rozowsky J, *Alexander R, *Min R, *Alves P, Abyzov A, Addleman N, Bhardwaj N, **Boyle AP**, Cayting P, Charos A, Chen DZ, Cheng Y, Clarke D, Eastman C, Euskirchen G, Frietze S, Fu Y, Gertz J, Grubert F, Harmanci A, Jain P, Kasowski M, Lacroute P, Leng J, Lian J, Monahan H, O’Geen H, Ouyang Z, Partridge EC, Patacsil D, Pauli F, Raha D, Ramirez L, Reddy TE, Reed B, Shi M, Slifer T, Wang J, Wu L, Yang X, Yip KY, Zilberman-Schapira G, Batzoglou S, Sidow A, Farnham PJ, Myers RM, Weissman SM, Snyder M. “Architecture of the human regulatory network derived from ENCODE data.” *Nature* 2012, 489(7414):91–100. PMID: 22955619.

[19] The ENCODE Project Consortium. “An integrated encyclopedia of DNA elements in the human genome.” *Nature* 2012, 489(7414):57–74. PMID: 22955616.

[18] **Boyle AP**, Hong EL, Hariharan M, Cheng Y, Schaub MA, Kasowski M, Karczewski KJ, Park J, Hitz BC, Weng S, Cherry JM, Snyder M. “Annotation of functional variation in personal genomes using RegulomeDB.” *Genome Research* 2012, 22(9):1790–1797. PMID: 22955989.

[17] *Chen R, *Mias GI, *Li-Pook-Than J, *Jiang L, Lam HYK, Chen R, Miriami E, Karczewski KJ, Hariharan M, Dewey FE, Cheng Y, Clark MJ, Im H, Habegger L, Balasubramanian S, O’Huallachain M, Dudley JT, Hillenmeyer S, Haraksingh R, Sharon D, Euskirchen G, Lacroute P, Bettinger K, **Boyle AP**, Kasowski M, Grubert F, Seki S, Garcia M, Whirl-Carrillo M, Gallardo M, Blasco MA, Greenberg PL, Snyder P, Klein TE, Altman RB,

Butte AJ, Ashley EA, Gerstein M, Nadeau KC, Tang H, Snyder M. “Personal omics profiling reveals dynamic molecular and medical phenotypes.” *Cell* 2012, 148(6):1293–1307. PMID: 22424236.

[16] *Song L, *Zhang Z, *Grasfeder LL, ***Boyle AP**, *Giresi PG, *Lee BK, *Sheffield NC, Graff S, Huss M, Keefe D, Liu Z, London D, McDaniell RM, Shibata Y, Showers KA, Simon JM, Vales T, Wang T, Winter D, Zhang Z, Clarke ND, †Birney E, †Iyer VR, †Crawford GE, †Lieb JD, †Furey TS. “Open chromatin defined by DNaseI and FAIRE identifies regulatory elements that shape cell-type identity.” *Genome Research* 2011, 21(10):1757–1767. PMID: 21750106.

[15] The ENCODE Project Consortium. “A User’s Guide to the Encyclopedia of DNA Elements (ENCODE).” *PLoS Biology* 2011, 9(4):e1001046. PMID: 21526222.

[14] **Boyle AP**, Song L, Lee BK, London D, Keefe D, Birney E, Iyer VR, †Crawford GE, †Furey TS. “High-resolution genome-wide in vivo footprinting of diverse transcription factors in human cells.” *Genome Research* 2011, 21(3):456–464. PMID: 21106903.

[13] Georgiev S, **Boyle AP**, Jayasurya K, Mukherjee S, Ohler U. “Evidence-ranked motif identification.” *Genome Biology* 2010, 11(2):R19. PMID: 20156354.

[12] *Stitzel ML, *Sethupathy P, Pearson DS, Chines PS, Song L, Erdos MR, Welch R, Parker SCJ, **Boyle AP**, Scott LJ, Margulies EH, Boehnke M, Furey TS, Crawford GE, Collins FS. “Global epigenomic analysis of primary human pancreatic islets provides insights into type 2 diabetes susceptibility loci.” *Cell Metabolism* 2010, 12(5):443–455. PMID: 21035756.

[11] McDaniell R, Lee BK, Song L, Liu Z, **Boyle AP**, Erdos MR, Scott LJ, Morken MA, Kucera KS, Battenhouse A, Keefe D, Collins FS, Willard HF, Lieb JD, Furey TS, †Crawford GE, †Iyer VR, †Birney E. “Heritable individual-specific and allele-specific chromatin signatures in humans.” *Science* 2010, 328(5975):235–239. PMID: 20299549.

[10] Babbitt CC, Fedrigo O, Pfefferle AD, **Boyle AP**, Horvath JE, Furey TS, Wray GA. “Both noncoding and protein-coding RNAs contribute to gene expression evolution in the primate brain.” *Genome Biology and Evolution* 2010, 2:67–79. PMID: 20333225.

[9] Xu X, Tsumagari K, Sowden J, Tawil R, **Boyle AP**, Song L, Furey TS, Crawford GE, Ehrlich M. “DNaseI hypersensitivity at gene-poor, FSH dystrophy-linked 4q35.2.” *Nucleic Acids Research* 2009, 37(22):7381–7393. PMID: 19820107.

[8] **Boyle AP**, Furey TS. “High-resolution mapping studies of chromatin and gene regulatory elements.” *Epigenomics* 2009, 1(2):319–329. PMID: 20514362.

[7] **Boyle AP**, Guinney J, Crawford GE, Furey TS. “F-Seq: a feature density estimator for high-throughput sequence tags.” *Bioinformatics* 2008, 24(21):2537–2538. PMID: 18784119.

[6] **Boyle AP**, Davis S, Shulha HP, Meltzer P, Margulies EH, Weng Z, †Furey TS, †Crawford GE. “High-resolution mapping and characterization of open chromatin across the genome.” *Cell* 2008, 132(2):311–322. PMID: 18243105.

[5] **Boyle AP**, Boyle JA, Bridges SM. “Identification of Regulatory Elements in Archaea using Self-Organizing Maps.” *Proceedings of the Conference on Research in Computational Molecular Biology* 2004, pp. 455–456. Publication link.

[4] **Boyle AP**, Boyle JA. “Global analysis of microbial translation initiation regions.” *Journal of the Mississippi Academy of Sciences* 2003, 48(3):138–150. Publication link.

[3] **Boyle AP**, Bridges S. “Clustering of archael gene regulatory regions.” *FASEB Journal* 2003, 17(5):A985.

[2] **Boyle AP**, Boyle JA. “Visualization of aligned genomic open reading frame data.” *Biochemistry and Molecular Biology Education* 2003, 31(1):64–68. doi: 10.1002/bmb.2003.494031010144.

[1] Wan X, Bridges SM, Boyle JA, **Boyle AP**. “Interactive clustering for exploration of genomic data.” *Proceedings of the Artificial Neural Networks in Engineering Conference* 2002, 12:753–758. Publication link.

Patents

[1] Karczewski K, Snyder M, Butte AJ, Dudley JT, Hong E, **Boyle A**, Cherry MJ, Park J. “Method and System for the Use of Biomarkers for Regulatory Dysfunction in Disease.” Granted United States patent 9,946,835, 2018. Patent record.